Mathe 1 MB Übungsblatt 5: Differenzialrechnung Lösungen

ABGABE VON
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(A 1)

1) $f(x) = 4x^3$

$f'(x) = 12x^2$

2) $f(x) = 4x^3 + x^2 + x + 1$

$f'(x) = 12x^2 + 2x + 1$

3) $f(x) = \sqrt{x} = x^{\frac{1}{2}}$

$f'(x) = \frac{1}{2} x^{-\frac{1}{2}} = \frac{1}{2} \frac{1}{x^{1/2}} = \frac{1}{2\sqrt{x}}$

4) $f(x) = \sqrt[4]{x^3} = (x^3)^{\frac{1}{4}} = x^{\frac{3}{4}}$

$f'(x) = \frac{3}{4} x^{-\frac{1}{4}} = \frac{3}{4} \frac{1}{x^{\frac{1}{4}}} = \frac{3}{4\sqrt[4]{x}}$

5) $f(t) = t^2 e^{3t}$

$f'(t) = 2te^{3t} + t^2 3e^{3t} = t \cdot e^{3t}(2 + 3t)$

6) $f(x) = (x^n + 1) \cdot \sqrt{x^2 - 1} = (x^n + 1)(x^2 - 1)^{\frac{1}{2}}$

$f'(x) = (n \cdot x^{n-1}) \cdot \sqrt{x^2 - 1} + (x^n + 1) \cdot \frac{1}{2} (x^2 - 1)^{-\frac{1}{2}} \cdot 2x$

$= (nx^{n-1}) \sqrt{x^2 - 1} + (x^n + 1) \cdot \frac{x}{\sqrt{x^2 - 1}}$

7) $f(x) = \frac{10x}{x^2 + 1} = 10x \cdot (x^2 + 1)^{-1}$

$f'(x) = 10 \cdot (x^2 + 1)^{-1} + 10x(-1) \cdot 2x (x^2 + 1)^{-2}$

$= \frac{10}{x^2 + 1} - \frac{20x^2}{(x^2 + 1)^2}$

$= \frac{10(x^2 + 1)}{(x^2 + 1)^2} - \frac{20x^2}{(x^2 + 1)^2}$

$= \frac{10x^2 + 10 - 20x^2}{(x^2 + 1)^2} = \frac{10(1 - x^2)}{(x^2 + 1)^2}$

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A1 (Fortsetzung)

⑧ $f(x) = \frac{1+x}{1+x^2} = \frac{g(x)}{h(x)}$

$$f'(x) = \frac{g'(x)\cdot h(x) - h'(x)\cdot g(x)}{h^2(x)}$$

$$= \frac{1\cdot(1+x^2) - 2x(1+x)}{(1+x^2)^2}$$

$$= \frac{1+x^2 - (2x+2x^2)}{(1+x^2)^2}$$

$$= \frac{-x^2-2x+1}{(1+x^2)^2}$$

9) $f(x) = 5(x^3-x^2+1)^5$

$$f'(x) = 25(x^3-x^2+1)^4(3x^2-2x)$$

10) $f(x) = \left(\frac{1+x}{x}\right)^n = \left((1+x)\cdot x^{-1}\right)^n$

$$f'(x) = n\cdot\left(\frac{1+x}{x}\right)^{n-1}\cdot\left(\frac{1+x}{x}\right)' \quad \text{(Kettenregel)}$$

$$= n\cdot\left(\frac{1+x}{x}\right)^{n-1}\cdot\frac{x\cdot 1 - 1\cdot(1+x)}{x^2} \quad \text{Quotientenregel}$$

$$= n\cdot\left(\frac{1+x}{x}\right)^{n-1}\cdot\frac{\cancel{x} - 1 - \cancel{x}}{x^2}$$

$$= -\frac{n}{x^2}\left(\frac{1+x}{x}\right)^{n-1}$$

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A1 (Fortsetzung)

11) $f(x) = 2x\sqrt{x^2-1} = \underbrace{2x}_{u}\ \underbrace{(x^2-1)^{\frac{1}{2}}}_{v}$ Produktregel

$$f'(x) = 2(x^2-1)^{\frac{1}{2}} + 2x\,\tfrac{1}{2}(x^2-1)^{-\frac{1}{2}}\cdot 2x$$

$$= 2\sqrt{x^2-1} + 2x^2\frac{1}{\sqrt{x^2-1}}$$

$$= 2\frac{x^2-1+2x^2}{\sqrt{x^2-1}} = 2\frac{3x^2-1}{\sqrt{x^2-1}}$$

12) $u(t) = u_0 \sin(\omega t + \varphi)$

$u'(t) = u_0\,\omega\cos(\omega t+\varphi)$

13) $f(x) = \tan(x) = \frac{\sin(x)}{\cos(x)}$

$$f'(x) = \frac{\cos(x)\cdot\cos(x) - \sin(x)\cdot(-\sin(x))}{\cos^2(x)}$$

$$= \frac{\cos^2(x)+\sin^2(x)}{\cos^2(x)} = \frac{1}{\cos^2(x)}$$ mit $\cos^2(x)+\sin^2(x)=1$

$$= \frac{\cos^2(x)}{\cos^2(x)} + \frac{\sin^2(x)}{\cos^2(x)}$$

$$= 1 + \tan^2(x)$$

14) $f(x) = a^x = \left(e^{\ln(a)}\right)^x = e^{x\ln(a)}$

$f'(x) = e^{x\ln(a)}\cdot\ln(a) = a^x\cdot\ln(a)$

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(A1) (Fortsetzung)

15) $f(x) = \sqrt{a^2 - x^2} = (a^2 - x^2)^{\frac{1}{2}}$

$f'(x) = \frac{1}{2}(a^2 - x^2)^{-\frac{1}{2}} \quad (-2x)$

$= \frac{-x}{\sqrt{a^2 - x^2}}$

16) $f(x) = \sin(2x)$

$f'(x) = 2\cos(2x)$

17) $f(x) = \frac{a}{x}\sqrt{a^2 + x^2} = \frac{a}{x}(a^2 + x^2)^{\frac{1}{2}}$ ← Produktregel

$f'(x) = -a x^{-2}(a^2 + x^2)^{\frac{1}{2}} + a x^{-1} \frac{1}{2}(a^2 + x^2)^{-\frac{1}{2}} \cdot 2x$

$= -\frac{a}{x^2}\sqrt{a^2 + x^2} + \frac{a}{x} \times \frac{1}{\sqrt{a^2 + x^2}}$

$= a\frac{-a^2 - x^2 + x^2}{\sqrt{a^2 + x^2}\, x^2} = -\frac{a^3}{x^2\sqrt{a^2 + x^2}}$

18) $f(x) = \frac{1}{\sin(x)} = (\sin(x))^{-1}$

$f'(x) = -(\sin(x))^{-2} \cdot \cos(x)$

$= -\frac{\cos(x)}{(\sin(x))^2}$

19) $f(x) = \cos^3(x) = (\cos(x))^3$

$f'(x) = 3(\cos(x))^2 \cdot (-\sin(x))$

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(A1) (Fortsetzung)

20) $f(x) = \sin(x^3)$

$f'(x) = \cos(x^3) \cdot 3x^2$

21) $f(x) = \tan^3(x) + 3\tan(x) = (\tan(x))^3 + 3\tan(x)$

$f'(x) = 3(\tan(x))^2 \cdot \frac{1}{\cos^2(x)} + \frac{3}{\cos(x^2)}$

$= \frac{3}{\cos(x^2)} \underbrace{(\tan(x)^2 + 1)}_{\frac{1}{\cos^2(x)}} = \frac{3}{\cos^4(x)}$

22) $f(x) = \tan\left(\frac{1}{x}\right)$

$f'(x) = \frac{1}{\cos^2\left(\frac{1}{x}\right)} \cdot \left(-\frac{1}{x^2}\right) = -\frac{1}{\cos^2(1/x) \cdot x^2}$

23) $f(x) = \frac{x^2 - a^2}{\sin(x)} = (x^2 - a^2)(\sin(x))^{-1}$

$f'(x) = 2x(\sin(x))^{-1} + (x^2 - a^2) \cdot (-1)(\sin(x))^{-2} \cdot \cos(x)$

$= \frac{2x}{\sin(x)} - \frac{x^2 - a^2}{(\sin(x))^2} \cdot \cos(x)$

24 $f(x) = \tan(x^2) \cdot x^2$

Produktregel ↙

$f'(x) = \left(\frac{1}{\cos(x^2)^2} \cdot 2x\right) \cdot x^2 + 2x \cdot \tan(x^2)$

$= 2x\left(\frac{x^2}{\cos^2(x^2)} + \tan(x^2)\right)$

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A7 (Fortsetzung)

25) $f(x) = 3x + \sin(\cos(x))$

$f'(x) = 3 + \cos(\cos(x)) \cdot (-1)\sin(x) = 3 - \sin(x) \cdot \cos(\cos(x))$

26) $f(x) = \sqrt[5]{x^4 - x^2} = (x^4 - x^2)^{\frac{1}{5}}$

$f'(x) = \frac{1}{5}(x^4 - x^2)^{-\frac{4}{5}} \cdot (4x^3 - 2x)$

$= \frac{1}{5} \cdot \frac{1}{\sqrt[5]{(x^4 - x^2)^4}} \cdot 2x(2x^2 - 1)$

27) $f(x) = \sqrt[5]{(x-6)^4}$

$f'(x) = \frac{4}{5}(x-6)^{\frac{4}{5}-1}$

$f'(x) = \frac{4}{5}(x-6)^{-\frac{1}{5}}$

$= \frac{4}{5} \cdot \frac{1}{\sqrt[5]{x-6}}$

28) $f(x) = e^{-x} \cdot x^2$

$f'(x) = e^{-x} \cdot 2x - e^{-x} x^2$

$= e^{-x} \cdot x(2-x)$

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(A2) a) $f(x) = ax + b$

$$f'(x) = \lim_{\Delta \to 0} \frac{a(x+\Delta x) + b - (ax+b)}{\Delta x}$$

$$= \lim_{\Delta \to 0} \frac{ax + a\Delta x + b - ax - b}{\Delta x}$$

$$= \lim_{\Delta \to 0} \frac{a \Delta x}{\Delta x} = a$$

b) $f(x) = \sqrt{x}$

$$f'(x) = \lim_{\Delta x \to 0} \left(\sqrt{x+\Delta x} - \sqrt{x}\right) \frac{\sqrt{x+\Delta x} + \sqrt{x}}{\sqrt{x+\Delta x} + \sqrt{x}}$$

$$= \lim_{\Delta x \to 0} \frac{x + \Delta x - x}{\left(\sqrt{x+\Delta x} + \sqrt{x}\right) \Delta x}$$

$$= \frac{1}{\sqrt{x} + \sqrt{x}} = \frac{1}{2\sqrt{x}}$$

(A3) $f(x) = (a-x)(a+x)$

a) $f'(x) = (a-x) \cdot 1 + (a+x) \cdot (-1)$ ← mit Produktregel

$= a - x - (a+x)$

$= a - x - a - x$

$= -2x$

b) $f(x) = a^2 - x^2$ ← ohne Produktregel

$f'(x) = -2x$

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(A4) $\frac{d}{dx} \arccos(x) = \frac{-1}{\sqrt{1-x^2}}$

$g(x) = \arccos(x)$ $\qquad$ $f(x) = \cos(x)$

$g'(x) = \frac{-1}{\sin(\arccos(x))}$ $\qquad$ $\sin^2(x) + \cos^2(x) = 1$

$\sin(x) = \sqrt{1-\cos^2(x)}$

$= \frac{-1}{\sqrt{1-\cos(\arccos(x))}} = \frac{-1}{\sqrt{1-x^2}}$

(A5) $f(x) = e^{-x} \cdot \sin(x)$

$f'(x) = -e^{-x} \cdot \sin(x) + \cos(x) \cdot e^{-x}$

$= e^{-x}(\cos(x) - \sin(x))$

$f''(x) = -e^{-x} \cdot (\cos(x) - \sin(x)) + e^{-x}(-\sin(x) - \cos(x))$

$= e^{-x}(-\cos(x) + \sin(x) + (-\sin(x) - \cos(x))$

$= e^{-x}(-2\cos(x)) = -2 \cdot \cos(x) \cdot e^{-x}$

$f'''(x) = 2e^{-x} \cdot \cos(x) + e^{-x} \cdot 2\sin(x)$

$= 2e^{-x} \cdot (\sin(x) + \cos(x))$

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(A7) $f(x) = \underbrace{\frac{x}{1+x^2}}_{v(x)} \cdot u(x)$

$u(x) = x \qquad v(x) = 1+x^2$

$u'(x) = 1 \qquad v'(x) = 2x$

$$f'(x) = \frac{1\cdot(1+x^2) - x\cdot 2x}{(1+x^2)^2}$$

$$= \frac{1+x^2-2x^2}{(1+x^2)^2}$$

$$= \frac{1-x^2}{(1+x^2)^2}$$

$$f'(x_0) = 0 \Rightarrow \frac{1-x_0^2}{(1+x^2)^2} = 0$$

$$1 - x_0^2 = 0 \qquad \boxed{x_0 = \pm 1}$$

$$f'(x) = \frac{1-x^2}{(1+x^2)^2}$$

$u(x) = 1-x^2$

$u'(x) = -2x$

$v(x) = (1+x^2)^2$

$= 2\cdot(1+x^2)\cdot 2x$

$v'(x) = 4x(1+x^2)$

$$f''(x) = \frac{-2x(1+x^2)^2 - (1-x^2)\cdot(4x)(1+x^2)}{(1+x^2)^4}$$

$$= \frac{-2x-2x^3-(4x-4x^3)}{(1+x^2)^3}$$

$$= \frac{-6x+2x^3}{(1+x^2)^3}$$

$$f''(-1) = \frac{-6\cdot(-1)+2\cdot(-1^3)}{(1+1^2)^3} = \frac{4}{8} = \frac{1}{2} > 0 \Rightarrow \text{Min}$$

$$f''(+1) = \frac{-6\cdot 1 + 2\cdot 1^3}{(1+1^2)^3} = -\frac{4}{8} = -\frac{1}{2} < 0 \Rightarrow \text{Max}$$

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(47) (Fortsetzung)

$f''(x) = 0$

$$\frac{-6x + 2x^3}{(1+x^2)^3} = 0$$

— kann nicht null werden

$-6x + 2x^3 = 0$

$x(2x^2 - 6) = 0$

$2x^2 - 6 = 0$

$2x^2 = 6$

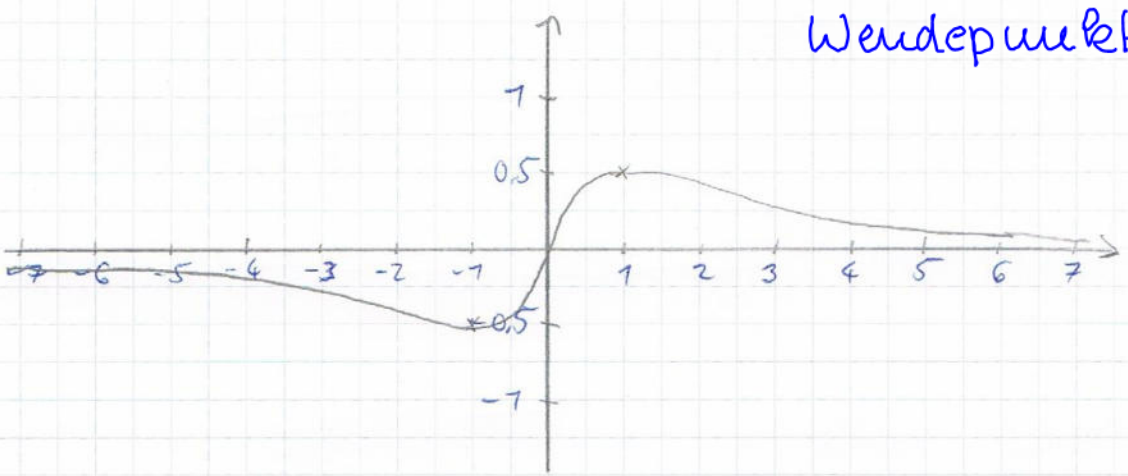
$x_1 = 0$

~~$x_2 = 0$~~

$x^2 = 3$

$x_{2/3} = \pm\sqrt{3}$

| hier sind Wendepunkte



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(AP) $f(x) = u(x)^{v(x)}$

$= \left(e^{\ln(u(x))}\right)^{v(x)}$

$= e^{vx \cdot \ln(u(x))}$

$f'(x) = e^{v(x)\ln(u(x))} \cdot \left(v'(x)\ln(u(x)) + v(x) \cdot \frac{u'(x)}{u(x)}\right)$

$$f'(x) = u(x)^{v(x)} \left(v'(x) \cdot \ln(u(x)) + \frac{v(x)u'(x)}{u(x)}\right)$$

a) $f(x) = x^x \qquad u(x) = x \qquad v(x) = x$

$f'(x) = x^x \left(1 \cdot \ln(x) + \frac{x \cdot 1}{x}\right)$

$= x^x (\ln(x) + 1)$

b) $f(x) = x^{-\ln(x)} \qquad u(x) = x \qquad v(x) = -\ln(x)$

$f'(x) = x^{-\ln(x)} \left(-\frac{1}{x} \cdot \ln(x)\right) + \frac{-\ln(x) \cdot 1}{x}$

$f'(x) = x^{-\ln(x)} \left(-\frac{2}{x} \cdot \ln(x)\right)$

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A9) $G(p) = G_0 (1+p)^n$

a) $G'(p) = G_0 \cdot n (1+p)^{n-1}$

$G'(0) = G_0 \cdot n (1)^{n-1}$

$= \underbrace{G_0 \cdot n}$

Steigung m der Tangente

$y = m\ p + c$

$= G_0 \cdot n \cdot p + c$

b) $y = 1000\,€ \cdot 5 \cdot 0{,}02 + 1000\,€$

$= 100\,€ + 1000\,€$

$= 1100\,€$

$G(0{,}02) = 1000\,€ (1 + 0{,}02)^5$

$= 1000\,€ (1{,}02)^5$

$= 1000\,€ \cdot 1{,}10408$

$= 1104{,}08\,€$

4,08€ Differenz, da bei der Tangente die Zinseszins nicht mitberechnet wird.

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A-10 $f(x) = 1 - e^{-x}(1+x)$

a)

$f'(x) = -e^{-x} + e^{-x}(1+x)$

$= e^{-x}(-1 + 1 + x) = x e^{-x}$

$f''(x) = e^{-x} + x(-e^{-x})$

$= e^{-x}(1-x)$

$f'''(x) = -e^{-x} + (-e^{-x})(1-x)$

$= -e^{-x}(1+1-x) = e^{-x}(x-2)$

b) $f''(x) = 0 \qquad 0 = e^{-x}(1-x)$

$x_1 = 1$

$f'''(x_1) = -e^{-1}(1+1-1)$

$0 \neq \frac{-1}{e}$ $\Rightarrow$ tatsächlich ein Wendepkt.

$f(1) = 1 - e^{-1}(1+1)$

$= 1 - \frac{1}{e}(2) = 1 - \frac{2}{e}$ ✓ $\Rightarrow$ $W(1 \mid 1 - \frac{2}{e})$

$t_{x_1}(x) = 1 - \frac{2}{e} + \frac{1}{e}(x-1)$ $\left(t_{x_1}(x) = f(x_1) + f'(x_1)(x - x_1)\right)$

$= 1 - \frac{2}{e} + x\frac{1}{e} - \frac{1}{e}$

$t_{x_1}(x) = \underbrace{1 - \frac{3}{e}}_{b} + x\underbrace{\frac{1}{e}}_{m}$

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A 10) (Fortsetzung)

c) $c = \lim_{x \to \infty} f(x)$

$c = \lim_{x \to \infty} \left(1 - e^{-x}(1+x)\right)$

$c = 1$

(x)=

d) $t_{x_1}(x) = 1 = c$

$1 - \frac{3}{e} + \frac{x}{e} = 1$

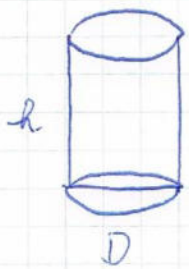
$\frac{x}{e} = \frac{3}{e} \qquad x = 3$

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A 11

$$D = \frac{4V}{\pi D^2} \qquad D^3 = \frac{4V}{\pi}$$

$$\boxed{D_0 = \sqrt[3]{\frac{4V}{\pi}}}$$



0,5 Liter = V

$V = \left(\frac{D}{2}\right)^2 \cdot \pi \cdot h$

$h = \frac{V4}{\pi D^2}$

$$u''(D) = \pi\left(\frac{4V}{\pi}\,\frac{2}{D_3} + 1\right)$$

$$u''(D_0) = \pi\left(\frac{8\cancel{V}}{\pi}\,\frac{\pi}{4\cancel{V}} + \cancel{1}\right)$$

$$= \pi \cdot (2+1) = 3\pi > 0$$

$\Rightarrow$ Minimum

$$h_0 = \frac{4V}{\pi D_0^2} = \frac{4V}{\pi\left(\frac{4V}{\pi}\right)^{\frac{2}{3}}}$$

$$u = D \cdot \pi \cdot h + \frac{D^2}{2} \cdot \pi$$

$$= D \cdot \pi\left(h + \frac{D}{2}\right)$$

$$u(D) = D \cdot \pi\left(\frac{4V}{\pi D^2} + \frac{D}{2}\right) = \pi\left(\frac{4V}{\pi D} + \frac{D^2}{2}\right)$$

$$u'(D) = \pi\left(\frac{4V}{\pi}\left(-\frac{1}{D^2}\right) + \frac{2D}{2}\right)$$

$$\overset{!}{0} \quad -\frac{4V}{\pi} \cdot \frac{1}{D^2} + \frac{2D}{2} = 0$$

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Aufgabe 11)

Volumen: $V = G \cdot h$ (G: Grundfläche, h: Höhe)



$V = (\pi R^2) \cdot h = \pi \left(\frac{D}{2}\right)^2 \cdot h$

$= \pi \frac{D^2}{4} \cdot h \quad \Rightarrow \quad \underline{\underline{h = \frac{4V}{\pi D^2}}}$

Mantelfläche

$M = 2 \cdot (\text{Unter/Oberseite}) + \text{Seitenfläche}$

$= 2 \cdot (\pi R^2) + h \cdot 2\pi R$

$= 2 \cdot \pi \frac{D^2}{4} + h \cdot 2 \cdot \pi \frac{D}{2} = \frac{\pi}{2} D^2 + \pi D \cdot h$

$= \pi \left(\frac{D^2}{2} + D \cdot h\right) = \pi \left(\frac{D^2}{2} + D \frac{4V}{\pi D^2}\right)$

$h = \frac{4V}{\pi D^2}$

$\underline{\underline{M(D) = \pi \left(\frac{D^2}{2} + \frac{4V}{\pi} \frac{1}{D}\right)}}$

Für welches D wird M(D) minimal?

$M'(D) = \pi \left(D - \frac{4V}{\pi} \frac{1}{D^2}\right)$

$M''(D) = 8V \cdot \frac{1}{D^3}$

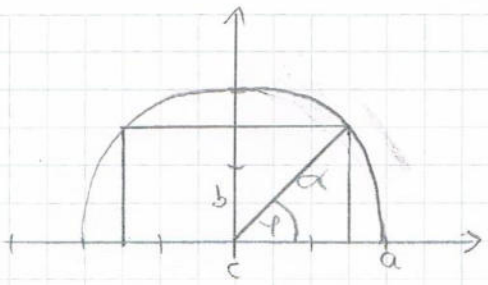
$M'(D_0) = 0 \quad \Rightarrow \quad D_0 = \frac{4V}{\pi} \frac{1}{D_0^2} \quad \Rightarrow \quad D_0^3 = \frac{4V}{\pi}$

$\underline{\underline{D_0 = \sqrt[3]{\frac{4V}{\pi}}}}$

$$M''(D_0) = 8V \frac{1}{D^3} = 8 \cdot V \frac{\pi}{4V} = 2\pi > 0$$

$\Rightarrow$ Minimum bei D_0

A12



$\frac{b}{a} = \sin(\alpha)$

$\frac{c/2}{a} = \cos(\alpha)$

$F = b \cdot c$

$F = (\sin(\alpha) \cdot a)(\cos(\alpha) \cdot 2a)$

$F(\alpha) = 2a^2 \cdot \sin(\alpha) \cdot \cos(\alpha)$

$F'(\alpha) = 2a^2\left((\sin(\alpha))(-\sin(\alpha)) + \cos(\alpha) \cdot \cos(\alpha)\right)$

$= 2a^2(\cos^2(\alpha) - \sin^2(\alpha))$

$0 = 2a^2(\cos^2(\alpha) - \sin^2(\alpha)$

$\cos^2(\alpha) = \sin^2(\alpha$

$\cos(\alpha) = \sin(\alpha)$

$\frac{\cos(\alpha)}{\cos(\alpha)} = \frac{\sin(\alpha)}{\cos(\alpha)}$

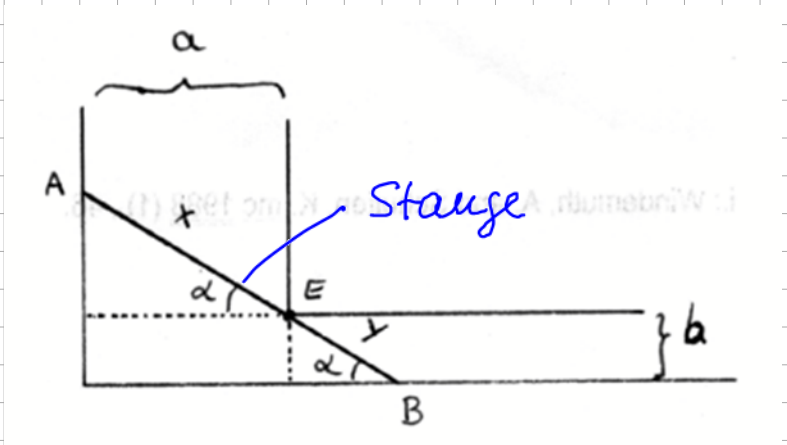
$1 = \tan(\alpha)$

$\arctan(1) = 45° = \alpha$

Für $\alpha = 45°$ wird die Fläche F maximal.

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Aufgabe 13)



$$\cos(\alpha) = \frac{a}{x}$$

$$\sin(\alpha) = \frac{b}{y}$$

Gesamtlänge der Stange: $L = x + y$

$$L = \frac{a}{\cos(\alpha)} + \frac{b}{\sin(\alpha)}$$

$$L(\alpha) = \frac{a}{\cos(\alpha)} + \frac{b}{\sin(\alpha)}$$

$$L'(\alpha) = -\frac{a \cdot \sin(\alpha)}{\cos^2(\alpha)} + \frac{b \cdot \cos(\alpha)}{\sin^2(\alpha)}$$

$$L''(\alpha) = -a \frac{\cos^2(\alpha) \cdot \cos(\alpha) - 2 \cdot \sin^2(\alpha) \cdot \cos(\alpha)}{(\cos^4(\alpha))^3}$$

$$+ b \frac{\sin^2(\alpha) \cdot (-\sin(\alpha)) - 2 \cdot \sin(\alpha) \cos^2(\alpha)}{(\sin^4(x))^3}$$

$$= -a \frac{\cos^2(\alpha) - 2\sin^2(\alpha)}{\cos^3(\alpha)} - b \frac{\sin^2(\alpha) + 2\cos^2(\alpha)}{\sin^3(\alpha)}$$

Nullstelle der 1. Ableitung

$$\Rightarrow \quad L'(\alpha_0) = 0 \quad \Rightarrow \quad -a \cdot \frac{\sin(\alpha_0)}{\cos^2(\alpha_0)} + b \frac{\cos(\alpha_0)}{\sin^2(\alpha_0)} = 0$$

$$\frac{b}{a} \frac{\cos(\alpha_0)}{\sin^2(\alpha_0)} = \frac{\sin(\alpha_0)}{\cos^2(\alpha_0)} \quad \Rightarrow \quad \frac{b}{a} = \left(\frac{\sin(\alpha_0)}{\cos(\alpha_0)}\right)^3$$

$$\tan^3(\alpha_0) = \frac{b}{a} \quad \Rightarrow \quad \tan(\alpha_0) = \sqrt[3]{\frac{b}{a}}$$

$$\alpha_0 = \arctan\left(\sqrt[3]{\frac{b}{a}}\right) = \arctan\left(\sqrt[3]{\frac{1}{2}}\right) \approx \underline{38{,}4}$$

Einsetzen von α_0 in 2. Ableitung

$L''(\alpha_0) = \ldots > 0 \quad \Rightarrow$ Minimum

$$L(\alpha_0) = \frac{8\,m}{\cos(\alpha_0)} + \frac{4m}{\sin(\alpha_0)}$$

$$\approx 1{,}02\,m + 6{,}44\,m \approx 7{,}46\,m$$

Die Stange darf nicht länger als 7,46 m sein, um durch die Gassen zu passen.

Aufgabe 14)

(a) $\lim\limits_{x \to 0} \frac{\sin(x)}{x} \quad \left(= \text{"}\frac{0}{0}\text{"} \right)$

$= \lim\limits_{x \to 0} \frac{(\sin(x))'}{x'} = \lim\limits_{x \to 0} \frac{\cos(x)}{1} = \frac{1}{1} = 1$

(b) $\lim\limits_{x \to 0} \frac{1}{x^2} \left(1 - \frac{1}{\cos(x)}\right) \quad \left(= \text{"}\frac{0}{0}\text{"} \right)$

$= \lim\limits_{x \to 0} \frac{\left(1 - \frac{1}{\cos(x)}\right)'}{x^2} = \lim\limits_{x \to 0} \frac{\frac{\sin(x)}{\cos^2(x)}}{2x} = \lim\limits_{x \to 0} \frac{-\sin(x)}{2x \cos^2(x)}$

$= \lim\limits_{x \to 0} \frac{(\sin(x))'}{(2x \cos^2(x))'} = \lim\limits_{x \to 0} \frac{\cos(x)}{2\cos^2(x) - 4x \cdot \sin(x) \cdot \cos(x)}$

$= +\frac{1}{2}$

(c) $\lim\limits_{x \to 0} \frac{\ln(2x+1)}{x} \quad \left(= \text{"}\frac{0}{0}\text{"} \right)$

$= \lim\limits_{x \to 0} \frac{(\ln(2x+1))'}{(x)'} = \lim\limits_{x \to 0} 2 \frac{1}{2x+1} = 2$

(d) $\lim\limits_{x \to 0} \frac{\arctan(x)}{x} \quad \left(= \text{"}\frac{0}{0}\text{"} \right)$

$= \lim\limits_{x \to 0} \frac{(\arctan(x))'}{(x)'} = \lim\limits_{x \to 0} \frac{1}{1+x^2} = 1$

(e) $\lim\limits_{x \to \infty} \frac{x^3}{e^x} \quad \left(= \text{"}\frac{\infty}{\infty}\text{"} \right)$

$= \lim\limits_{x \to \infty} \frac{3x^2}{e^x} = \lim\limits_{x \to \infty} \frac{6x}{e^x} = \lim\limits_{x \to \infty} \frac{6}{e^x} = \frac{6}{\infty} = 0$